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#include <iostream>
#include <vector>
using namespace std;

#define INF 9999

int main() {
int n;
cout << "Enter number of routers: ";
cin >> n;

vector<vector<int>> cost(n, vector<int>(n));
vector<vector<int>> dist(n, vector<int>(n));
vector<vector<int>> nextHop(n, vector<int>(n));

cout << "Enter cost matrix (9999 for no direct connection):\n";

for (int i = 0; i < n; i++) {
   for (int j = 0; j < n; j++) {
      cin >> cost[i][j];
      dist[i][j] = cost[i][j];

      if (i != j && cost[i][j] != INF)
         nextHop[i][j] = j;
      else
         nextHop[i][j] = -1;
   }
}

// Distance Vector Routing
for (int k = 0; k < n; k++) {
   for (int i = 0; i < n; i++) {
      for (int j = 0; j < n; j++) {
         if (dist[i][k] != INF &&
            dist[k][j] != INF &&
            dist[i][k] + dist[k][j] < dist[i][j]) {

            dist[i][j] = dist[i][k] + dist[k][j];
            nextHop[i][j] = nextHop[i][k];
         }
      }
   }
}

cout << "\nShortest Distance and Path:\n";

for (int i = 0; i < n; i++) {
```

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    for (int j = 0; j < n; j++) {
        if (i == j)
            continue;

        cout << "\nRouter " << i + 1
             << " to Router " << j + 1 << ":\n";

        if (dist[i][j] == INF) {
            cout << "Destination unreachable\n";
            continue;
        }

        cout << "Shortest distance: " << dist[i][j] << endl;
        cout << "Path: " << i + 1;

        int current = i;
        int count = 0;

        while (current != j && count < n) {
            current = nextHop[current][j];

            if (current == -1)
                break;

            cout << " -> " << current + 1;
            count++;
        }

        cout << endl;
    }
}

return 0;

}
```